

A Comparative Analysis of Classical Machine Learning Methods and Variational Quantum Methods for Breast Tumor Classification

Charisis Nikolaos

National and Kapodistrian University of Athens, Department of Informatics and Telecommunications

February 23, 2026

Outline

- 1 Classical Machine Learning Introduction
- 2 Quantum Computing and Quantum Machine Learning Basics
 - Quantum Mechanics and Quantum Computers
 - The Qubit
 - Qubit Manipulation
 - Manipulation of Multiple Qubits
 - Quantum Entanglement
 - Quantum Gates
 - Input Encoding Techniques
 - Variational Quantum Classifiers
- 3 Results from Classical Machine Learning methods
- 4 Results from Variational Quantum methods
- 5 Comparison & Conclusion
 - Comparison
 - Conclusion

Our goals:

- Tumor Classification using Classical ML methods

- Tumor Classification using Variational Quantum methods

- Comparison of the results

Workflow of the master thesis:

- Classical Machine Learning Methods Introduction

- Quantum Computing and Quantum Machine Learning Basics

- EDA & Classification using Classical ML methods and rnCV

- Classification using PCA and VQCs

- Comparison of the Classical and Quantum approaches

Classical Machine Learning Basics

Two types of ML:

- Supervised (Labeled Datasets)

- Unsupervised (Hidden patterns, no labels)

Unsupervised ML:

- Clustering

- Anomaly Detection

- Dimensionality Reduction

Supervised ML:

- Regression

- Classification

We will focus on **Classification**

Binary Classification with Classical ML Methods

Binary Classification: Choose between **two categories**

Classical ML methods:

- Logistic Regression**

- Gaussian Naive Bayes (GNB)**

- Linear Discriminant Analysis (LDA)**

- Support Vector Machines (SVM)**

- Random Forests**

- Light Gradient Boosting Machine (LightGBM)**

Quantum Computing and Quantum Machine Learning Basics

Key points:

- The Qubit

- Qubit Manipulation and Quantum Entanglement

- Quantum Gates

- Quantum Circuits

- Data Encoding

- Variational Quantum Classifiers

Quantum Mechanics and Quantum Computers

Quantum Theory or Quantum Mechanics:

Mathematical **Framework** for the results of measurement of **quantum systems**

Quantum Systems:

Microscopic entities where **Newtonian** mechanics **fail** (e.g hydrogen atom)

Quantum Computer:

Computational device that **exploits quantum mechanical effects** (superposition, entanglement) to carry out **operations on information**

Uses **Qubits** instead of bits

Quantum characteristics of particles can be **utilized**

Quantum processes can **manipulate** this information

The Qubit

Fundamental building blocks of quantum computers

The quantum mechanical analogue of a classical bit

Created by manipulating and measuring systems that exhibit quantum mechanical behavior

Superconducting qubits: **superconducting** materials at **cryogenic** temperatures, offer **rapid** gate speeds and **precise** control

Trapped ion qubits: individual **ions confined** by electromagnetic fields, **long coherence** times and **high-fidelity measurements**, slower than **superconducting**

Quantum dots: tiny **semiconductors** capture a **single electron**, potential for **scalability**

Photons: particles of **light**, encode quantum information and transmit via **optical fibers**, applications in quantum **communication** and **cryptography**

The Qubit, more formally

A two-state quantum system defined by the **basis vectors** $|0\rangle$ and $|1\rangle$

A qubit may exist in a **superposition**, a mathematical **linear combination** of both basis states, $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$

$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, α, β probability amplitudes with complex values

Hermitian conjugate: $\langle\psi| = (\alpha^* \quad \beta^*)$

$|\cdot\rangle$ is called "**ket**", $\langle\cdot|$ is called "**bra**"

Upon **measurement**, Qubit **collapses** with $P(|0\rangle) = |\alpha|^2$ and $P(|1\rangle) = |\beta|^2$

Thus, it must hold $|\alpha|^2 + |\beta|^2 = 1$ or $\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1$

If not, then we do $|\tilde{\psi}\rangle = \frac{|\psi\rangle}{\|\psi\|}$

A qubit is defined as a **vector** residing in a **two-dimensional complex Hilbert** space $\mathbb{C}^2$

Bloch sphere

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

$|\psi\rangle = |0\rangle$ occurs at $\theta = 0$

$|\psi\rangle = |1\rangle$ occurs at $\theta = \pi$

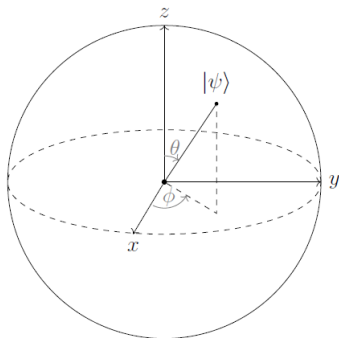


Figure: A qubit can be mapped to a point on the surface of this unit-radius sphere ($r = 1$), uniquely determined by the angular coordinates θ and ϕ

We transition qubits through **different states** with the use of **operators**

$$\hat{A}|\psi\rangle = |\phi\rangle, \hat{A}\langle\psi| = \langle\phi|$$

Vital set of operators called **Pauli** operators:

$$\hat{I}|\psi\rangle = |\psi\rangle$$

$$\hat{X}|0\rangle = |1\rangle, \hat{X}|1\rangle = |0\rangle \text{ (NOT operator)}$$

$$\hat{Y}|0\rangle = -i|1\rangle, \hat{Y}|1\rangle = i|0\rangle$$

$$\hat{Z}|0\rangle = |0\rangle, \hat{Z}|1\rangle = -|1\rangle$$

We can also express an operator as an outer product

$$(|\psi\rangle\langle\phi|)|\chi\rangle = |\psi\rangle\langle\phi|\chi\rangle = \langle\phi|\chi\rangle|\psi\rangle$$

Operators as Matrices

Another method for representing an operator is through matrix notation

For an n -**dimensional** vector space $\rightarrow n \times n$ **matrices**. We focus on $\mathbb{C}^2$, so 2×2 **matrices**

$$\hat{A} = \begin{pmatrix} \langle 0 | \hat{A} | 0 \rangle & \langle 0 | \hat{A} | 1 \rangle \\ \langle 1 | \hat{A} | 0 \rangle & \langle 1 | \hat{A} | 1 \rangle \end{pmatrix} \quad (1)$$

Pauli Operators as Matrices

Following Equation 1 we get that:

$$\hat{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (2)$$

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (3)$$

$$\hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (4)$$

$$\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (5)$$

Hermitian & Unitary Operators

A given operator $\hat{A}$ is classified as **Hermitian** iff:

$$\hat{A} = \hat{A}^\dagger \quad (6)$$

$\hat{A}^\dagger$ denotes the Hermitian adjoint $\rightarrow$ transposing the matrix and taking the complex conjugate of every entry

Hermitians possess **real eigenvalues** and their **main diagonal** consists solely of **real** numbers.

Therefore they represent **physical observables**

A given operator $\hat{U}$ is classified as **Unitary** iff:

$$\hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = \hat{I} \quad (7)$$

They characterize the **temporal evolution** of a quantum state

The **Pauli** operators are **both** the Hermitian and Unitary

To explore more sophisticated and advantageous quantum algorithms, we need systems of **multiple interacting qubits**

Construction of a **composite** Hilbert space

The mathematical operation used is the Kronecker or tensor product ($\otimes$)

If H_1 and H_2 two Hilbert spaces with dimensions N_1 and N_2 , then we can form a composite Hilbert space H with dimension $N_1 N_2$, defined as:

$$H = H_1 \otimes H_2 \quad (8)$$

States derived from tensor product

The tensor product can be employed to construct a state residing within the composite space H :

$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle \quad (9)$$

For two-dimensional subspaces, the following procedure is employed to calculate the product:

$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix} \quad (10)$$

Usually, the $\otimes$ symbol is omitted; for example, $|\phi\rangle \otimes |\chi\rangle$ may be written as $|\phi\rangle |\chi\rangle$, or even more compactly as $|\phi\chi\rangle$.

Basis vectors and operators of a composite system

If $|u_i\rangle$ and $|v_i\rangle$ are the basis vectors of H_1 and H_2 , the resulting basis for the joint space H is:

$$|w_i\rangle = |u_i\rangle \otimes |v_i\rangle \quad (11)$$

We can define operators that act upon our composite system. If $|\phi\rangle \in H_1$ and $|\chi\rangle \in H_2$, and A acting on $|\phi\rangle$ and B acting on $|\chi\rangle$. A new operator, which acts on a state $|\psi\rangle \in H$ is:

$$(A \otimes B) |\psi\rangle = (A \otimes B)(|\phi\rangle \otimes |\chi\rangle) = (A|\phi\rangle) \otimes (B|\chi\rangle) \quad (12)$$

The matrix representation is computed as:

$$A \otimes B = \begin{pmatrix} \alpha_{11}B & \alpha_{12}B \\ \alpha_{21}B & \alpha_{22}B \end{pmatrix} = \begin{pmatrix} \alpha_{11}\beta_{11} & \alpha_{11}\beta_{12} & \alpha_{12}\beta_{11} & \alpha_{12}\beta_{12} \\ \alpha_{11}\beta_{21} & \alpha_{11}\beta_{22} & \alpha_{12}\beta_{21} & \alpha_{12}\beta_{22} \\ \alpha_{21}\beta_{11} & \alpha_{21}\beta_{12} & \alpha_{22}\beta_{11} & \alpha_{22}\beta_{12} \\ \alpha_{21}\beta_{21} & \alpha_{21}\beta_{22} & \alpha_{22}\beta_{21} & \alpha_{22}\beta_{22} \end{pmatrix} \quad (13)$$

Quantum Entanglement

A composite system **cannot always** be expressed as a tensor product of individual states, only if the qubits are prepared separately and kept in isolation. If the qubits are permitted to **interact**, they form a single integrated closed system and are described as **entangled**:

Quantum Entanglement is a fundamental phenomenon and a vital resource for quantum computation

Particles sharing a common origin remain fundamentally **linked** regardless of the distance between them

Any measurement or interaction of one particle instantaneously **influences** the state of its entangled **partners**

Information can be transmitted **faster** than the speed of **light**

Quantum Gates

Transformations applied to qubit states are referred to as quantum gates, and a sequence of such operations forms a **quantum circuit**:

Quantum gates represent unitary **operations** performed on **qubits**

The terms "gate" and "unitary operator" are used **interchangeably**

A gate processing n **inputs** and outputs is represented by a $2^n \times 2^n$ **matrix**

A **single-qubit** gate is depicted as a 2×2 **matrix**

A quantum gate acting on **two qubits** requires a 4×4 **matrix** representation

The most fundamental single-qubit operation is the quantum *NOT* gate, which is equivalent to the Pauli X operator. This gate effectively swaps the basis states. Represented as a matrix in the computational basis, we have:

$$\hat{U}_{NOT} = \hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (14)$$

Hadamard and Rotation gates

The Hadamard gate, which transforms the standard basis states into a balanced superposition

$$\hat{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (15)$$

Rotations of a qubit along the x, y, and z axes of the Bloch sphere:

$$\hat{R}_x(\theta) = e^{-i\theta\frac{X}{2}} = \begin{pmatrix} \cos(\frac{\theta}{2}) & -i\sin(\frac{\theta}{2}) \\ -i\sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix} \quad (16)$$

$$\hat{R}_y(\theta) = e^{-i\theta\frac{Y}{2}} = \begin{pmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix} \quad (17)$$

$$\hat{R}_z(\theta) = e^{-i\theta\frac{Z}{2}} = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{-i\frac{\theta}{2}} \end{pmatrix} \quad (18)$$

Any arbitrary single-qubit gate can be represented as a specific sequence of rotations around the z and y axes (Z-Y decomposition)

The *controlled-NOT* (*CNOT*) gate

Two-qubit gate where the target qubit remains unchanged if the control qubit is in the state $|0\rangle$; however, if the control qubit is in the state $|1\rangle$, the target qubit undergoes a bit-flip, as if a *NOT* gate had been applied

$$CNOT|00\rangle = |00\rangle$$

$$CNOT|01\rangle = |01\rangle$$

$$CNOT|10\rangle = |11\rangle$$

$$CNOT|11\rangle = |10\rangle$$

Its matrix representation is :

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (19)$$

Quantum Circuits

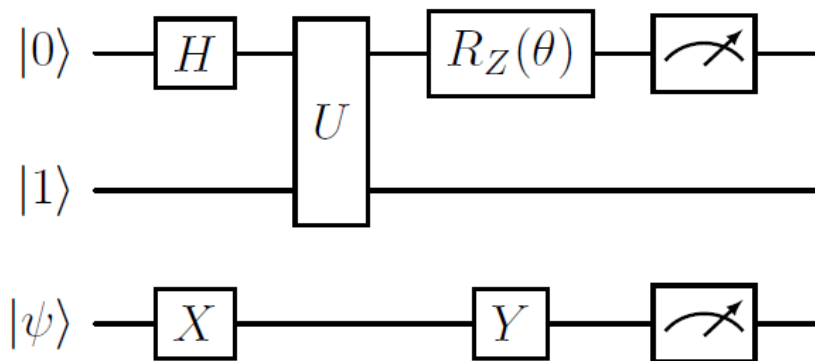


Figure: An example of a quantum circuit diagram

Input Encoding Techniques

Various techniques to **embed classical data** into an n -qubit quantum system:

- Basis** encoding

- Time-evolution** encoding

- Hamiltonian** encoding

- Amplitude** encoding

We will be using **Amplitude** encoding with **Amplitude-Efficient State Preparation** to encode 2^n features using n qubits

Variational Quantum Classifiers

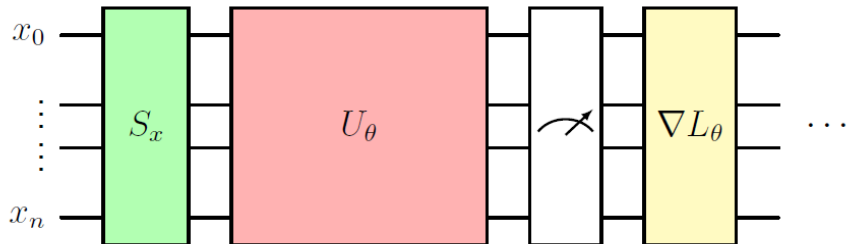


Figure: General structure of a variational quantum classifier

Breast Cancer

Most common malignancy among women worldwide

36% of all oncological cases

An estimated **2.089 million** women diagnosed in 2018

Lifestyle related factors:

Unhealthy dietary habits

Nicotine use

Chronic **stress**

Low levels of **physical** activity

Other risk factors:

Advanced **age**

Family **history** of breast cancer

Infection with **oncogenic** viruses

Early onset of **menstruation** or **late menopause**

First pregnancy at an **advanced** age or the **absence** of pregnancy

Use of **hormone replacement therapy**

Fine Needle Aspiration

We will be working with samples derived from Fine Needle Aspirates

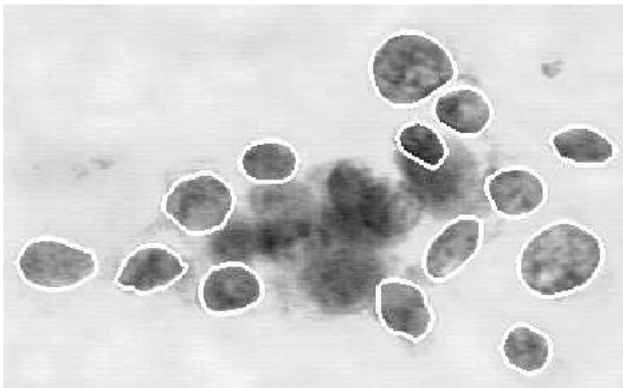


Figure: A magnified image of a malignant breast FNA with the visible cell nuclei having been outlined with the help of a curve fitting program.

Evaluation Metrics

Six models from before with **nine** key **metrics**:

$$\text{(Balanced) Accuracy } \left(\frac{1}{2} \left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP} \right) \right)$$

$$\text{Precision } \left(\frac{TP}{TP+FP} \right)$$

$$\text{Recall } \left(\frac{TP}{TP+FN} \right)$$

$$\text{F1-Score } \left(2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP+FP+FN} \right)$$

$$\text{MCC } \left(\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \right)$$

$$\text{ROC AUC } \left(\int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR}) \right)$$

$$\text{PR AUC } \left(\int_0^1 \text{Precision}(\text{Recall}) d(\text{Recall}) \right)$$

$$\text{Specificity } \left(\frac{TN}{TN+FP} \right)$$

$$\text{NPV } \left(\frac{TN}{TN+FN} \right)$$

Exploratory Data Analysis (feature distribution, class imbalance)

Finding the **best** model (80-20 split, $rnCV$, data cleaning)

Fine tune the winner (bootstrapping, 95% confidence intervals)

Exploratory Data Analysis

30 features with **569** entries

Class **imbalance**: **357** instances of **benign**, **212** instances of **malignant** (62.74% - 37.25%)

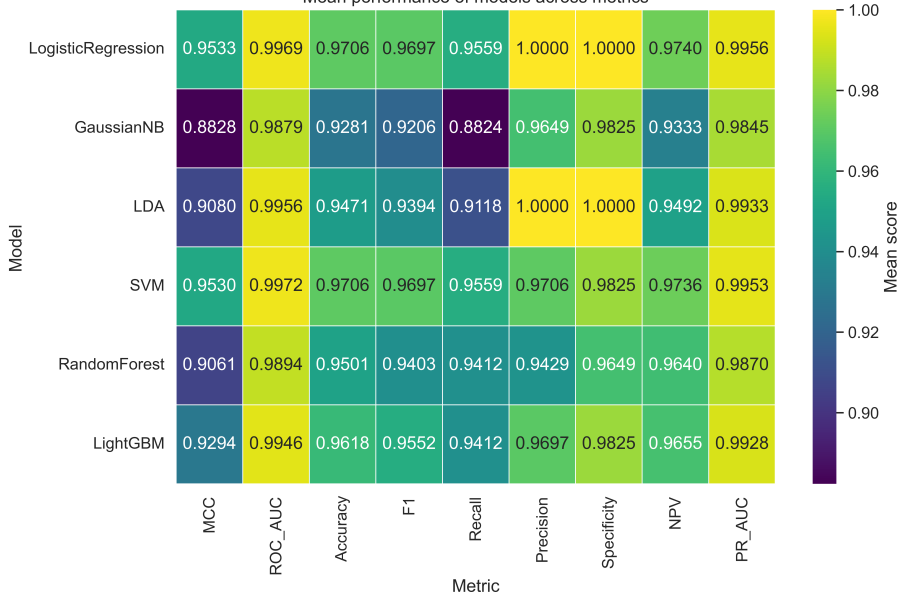
Majority of features exist on **different** scales, will **scale inside** rnCV

15 out of 30 features heavily **correlated** with **prognosis**

No correlation among features that we did not expect

Logistic Regression as the best model

Mean performance of models across metrics



Finetuning and bootstrapping

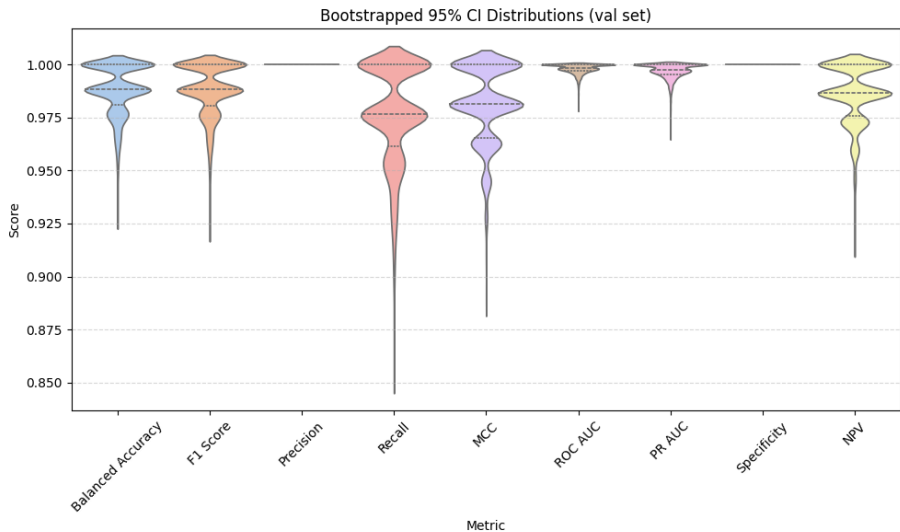


Figure: Violin plot of the confidence intervals

Conclusion

VQCs show strong potential, especially in low dimensionality problems

Classical ML still slightly superior

Future quantum hardware will improve results

Thank you for your attention!

Questions?